

MATHify.

Kids don't hate maths. They hate memorising it.

A gamified multiplication platform for children aged 7+ adaptive lessons, timed challenges, XP, ranks, badges and streaks.

Flutter

Node.js

MongoDB

Case study



Three reasons multiplication drills fail

01

One difficulty for every child

A 7 year-old and a 12 year-old get the same worksheet.

02

A wrong answer teaches nothing

A red cross tells a child they failed, not why.

03

No reason to come back tomorrow

Practice stops the moment the homework does.

Four pillars, twelve features

LEARN

Adaptive lessons

Lesson paths, an interactive engine and four hint styles per fact.

PLAY

Challenge mode

Six timed tiers, combo multipliers and scored results.

LEVEL UP

XP, ranks, badges

Rank progression, unlockable badges and collectible avatars.

COME BACK

Streaks & ranking

Daily streaks, reminders and a live leaderboard.

One screen tells a kid where they stand

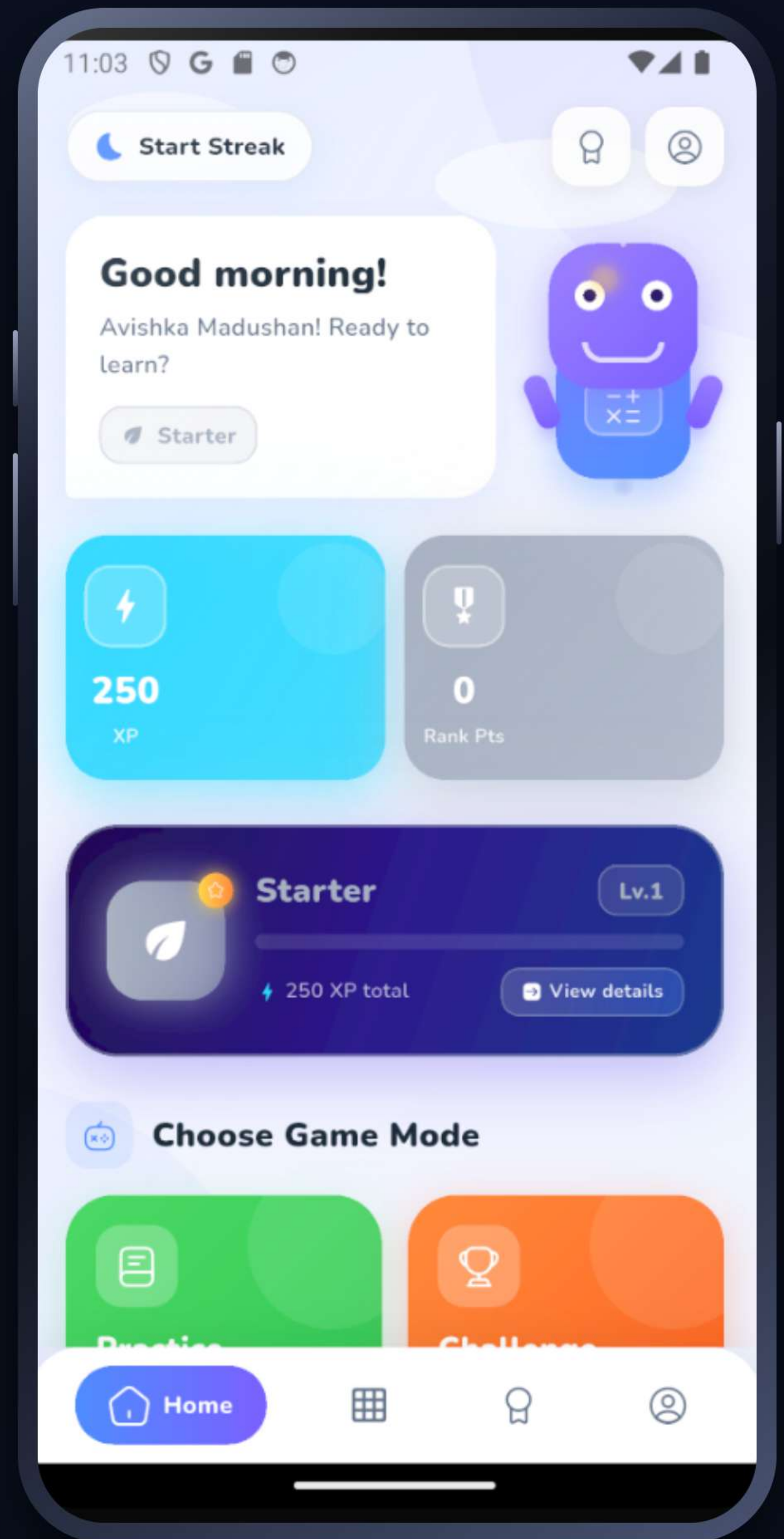
XP, rank points, current rank and level, streak status and the two game modes no menus, no reading required.

Greeting by time of day

Live XP + RP

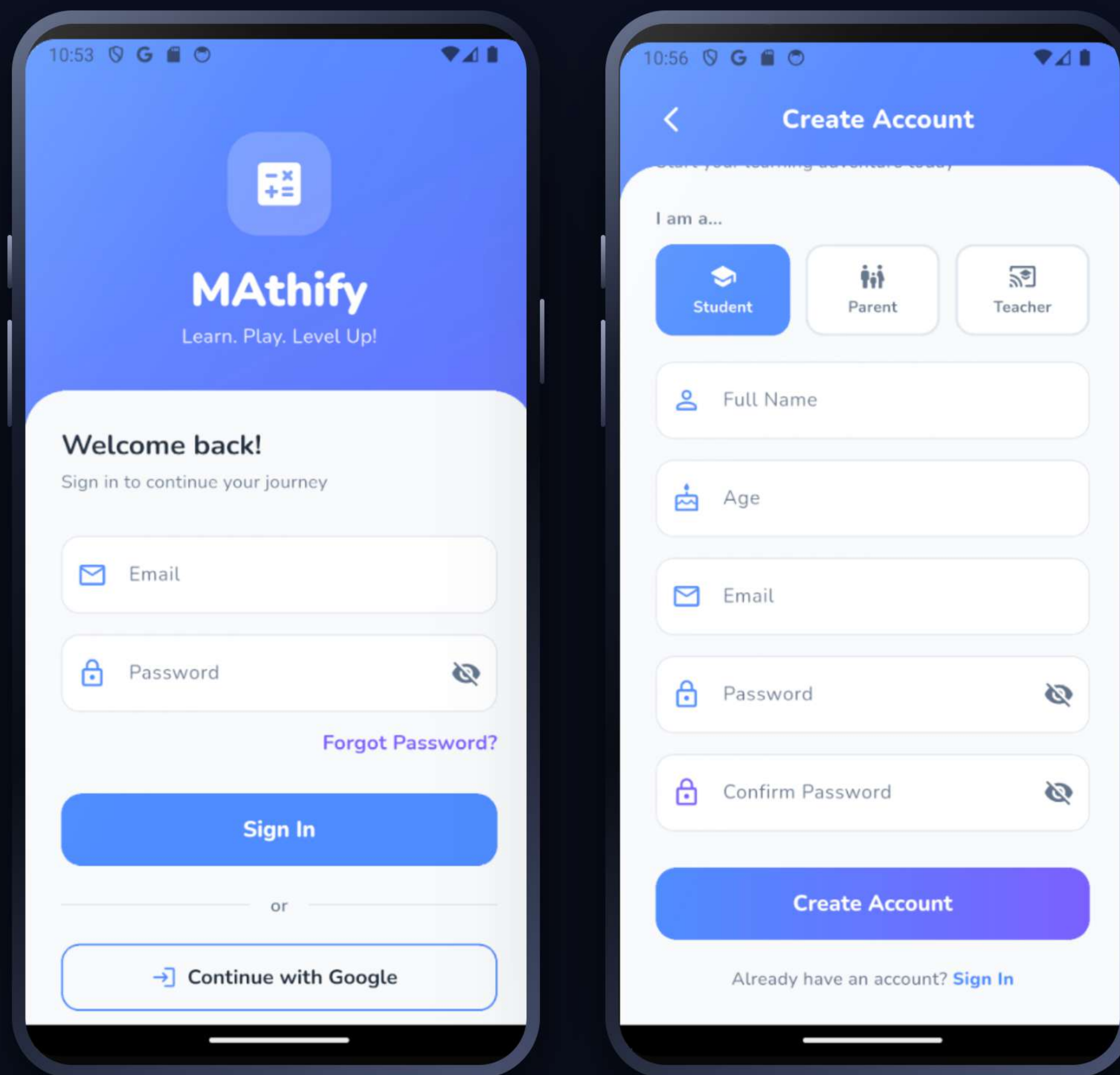
Rank card

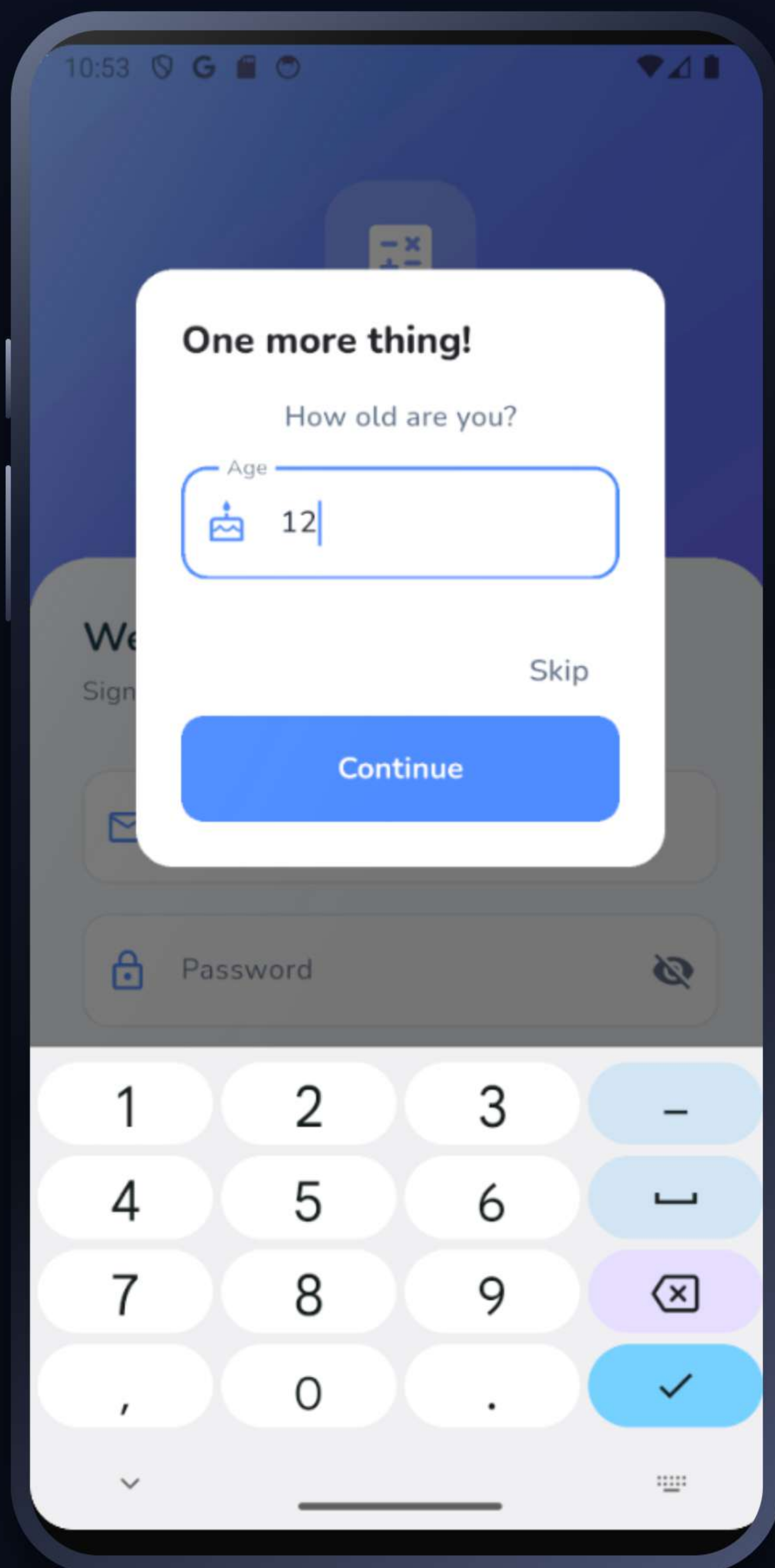
Streak CTA



Sign in a child can actually complete

Email + password, Google Sign In, and a role picker for Student, Parent or Teacher. Tokens held in secure storage.





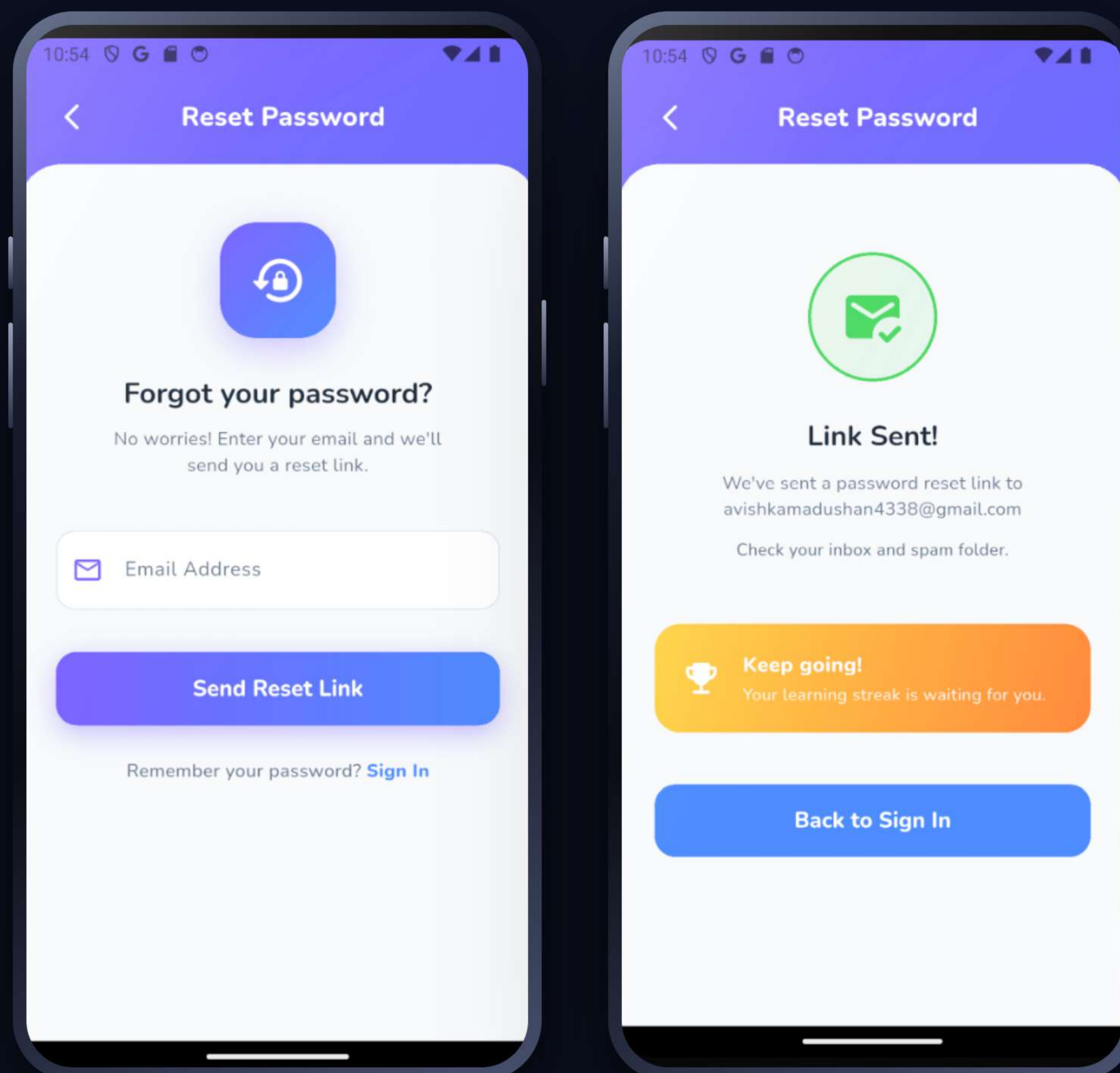
One question. The whole app adapts.

Age sets the starting table range, question count and timer length so a 7-year-old and a 12-year-old never see the same first lesson. Skippable, with a numeric keypad and no typing.

- Age 7–8 → tables 1–5, generous timers
- Age 9–10 → tables 1–9
- Age 11+ → tables 1–12, timed tiers

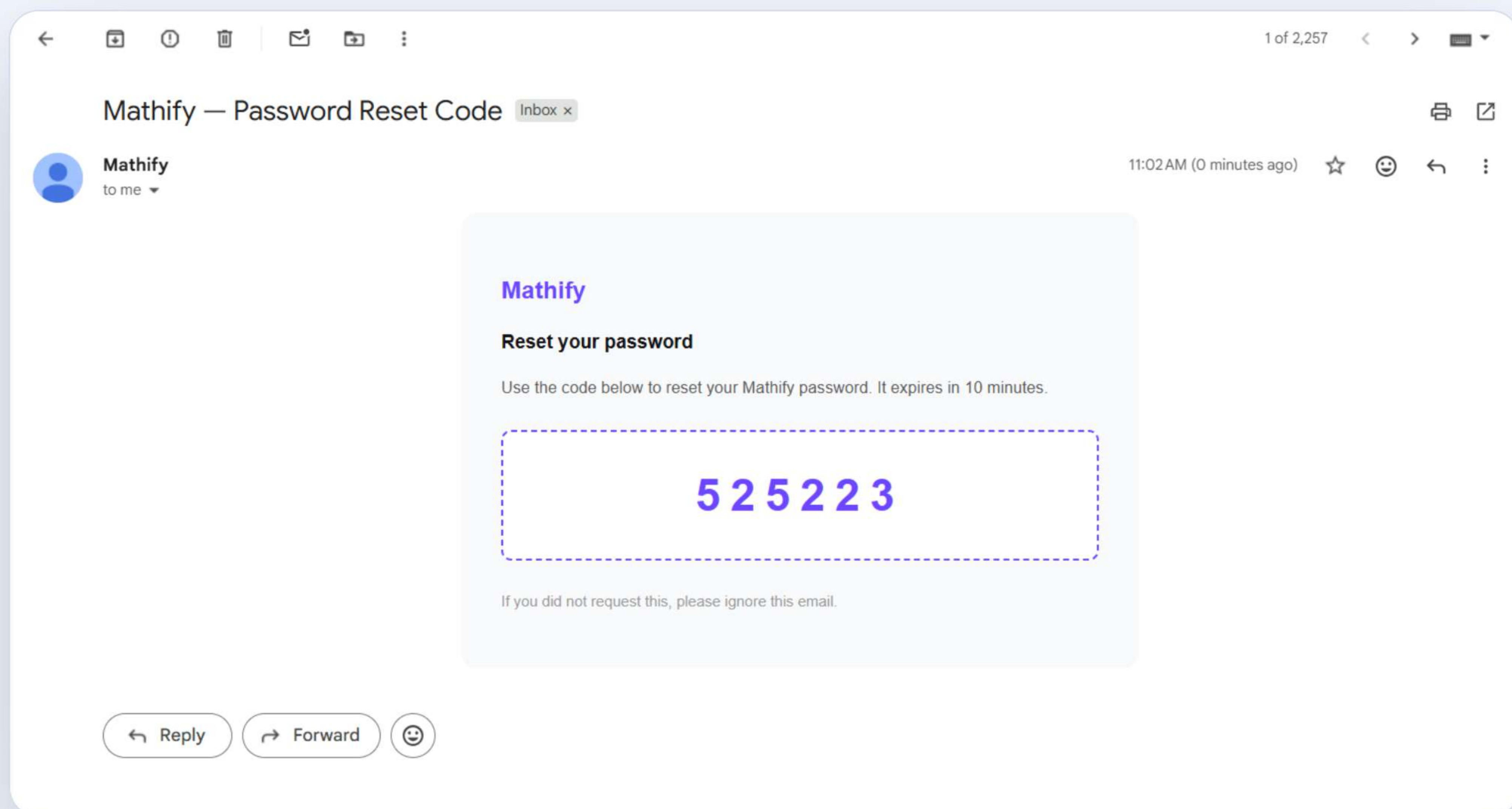
Forgot password — the boring flow, done properly

Request, confirmation, inbox hint, and a motivational nudge so the child comes back to their streak.



Transactional email, designed — not default

Node.js sends a branded HTML reset code with a 10-minute expiry. Same template, tested on desktop and mobile clients.



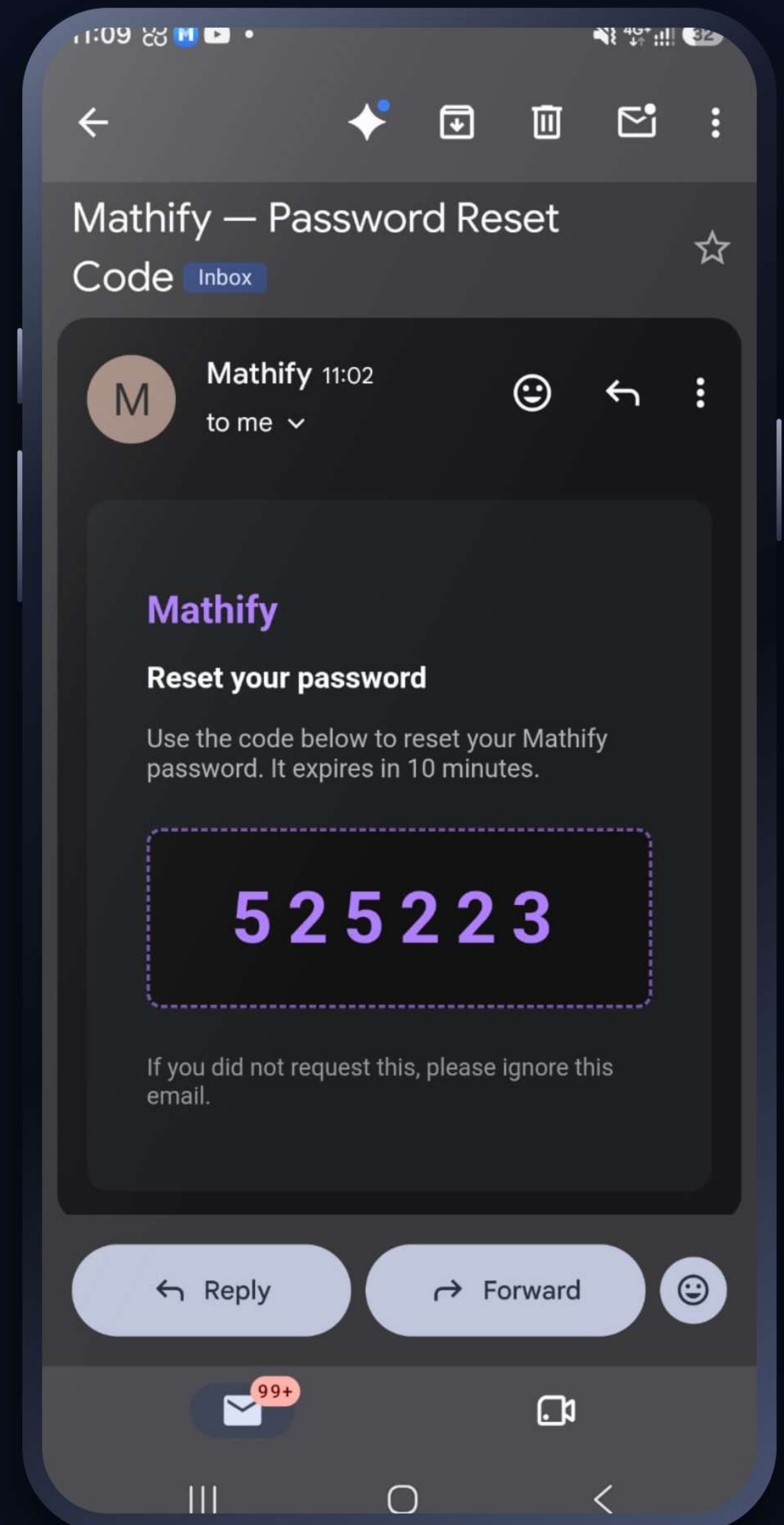
The same email, on a dark-mode phone

Tested on a real device in a real inbox: the code stays legible, the accent survives colour inversion, and nothing depends on a white background.

Real-device QA

Colour-scheme safe

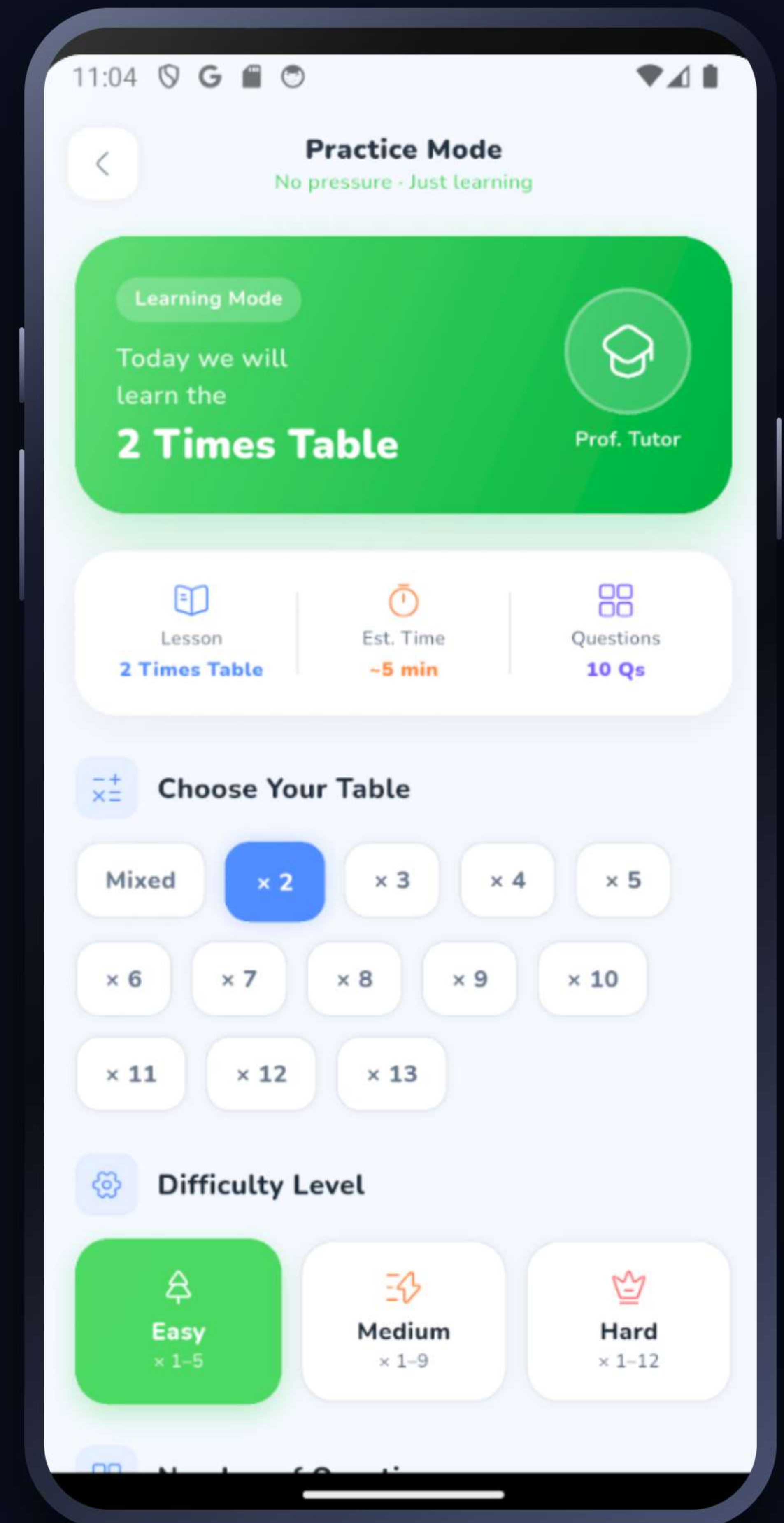
10-min expiry

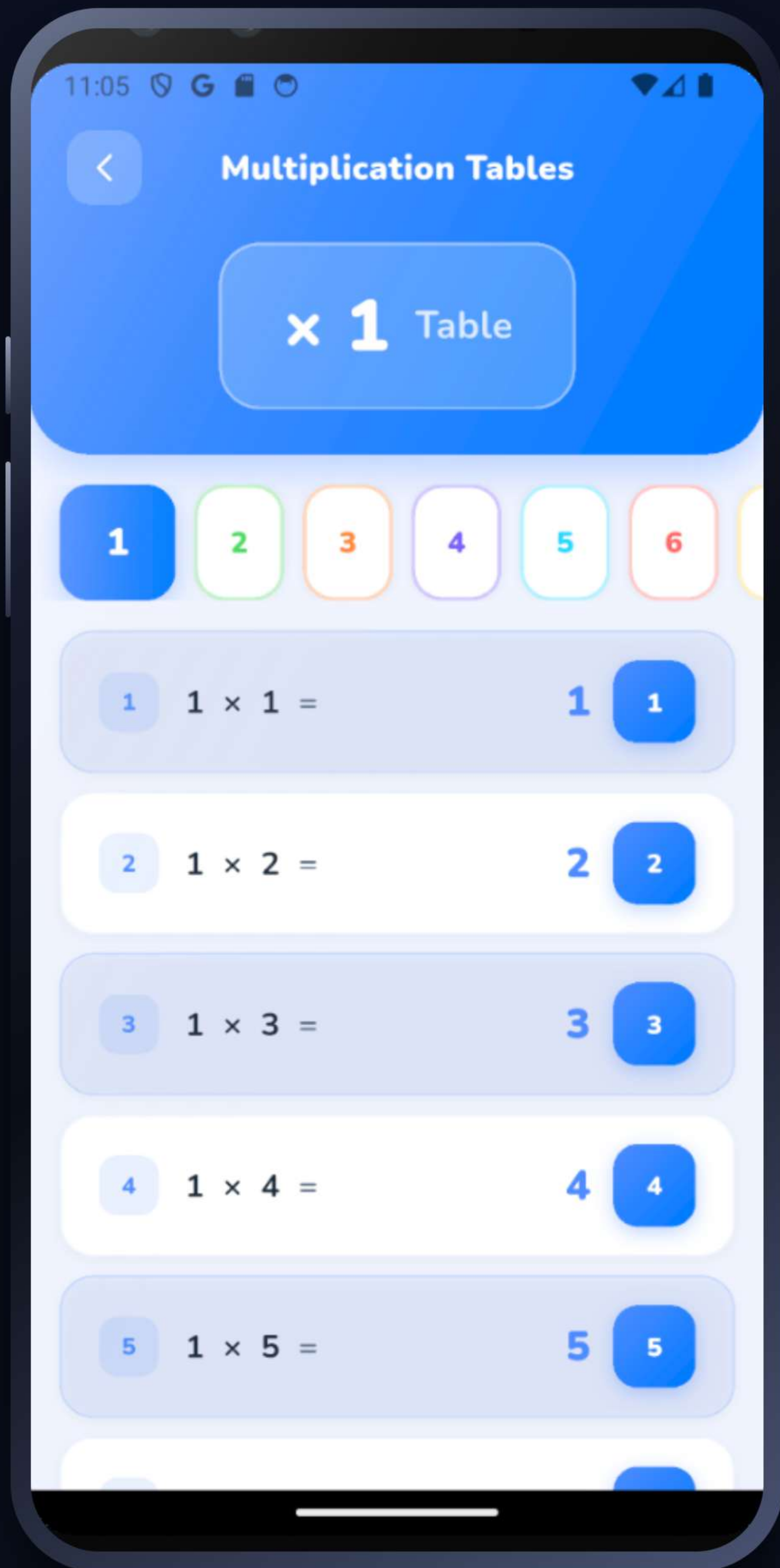


No timer. No pressure. Just learning.

Practice mode lets a child pick the table, the difficulty and how many questions then shows the estimated time before they start, so nothing feels like a test.

- 13 tables plus a mixed set
- Easy / Medium / Hard ranges
- Lesson, time estimate and question count up front





Times tables, always one tap away

A colour-coded reference for every table from 1 to 13, with a horizontal picker and zebra rows so a child can scan a single line without losing their place.

13 tables

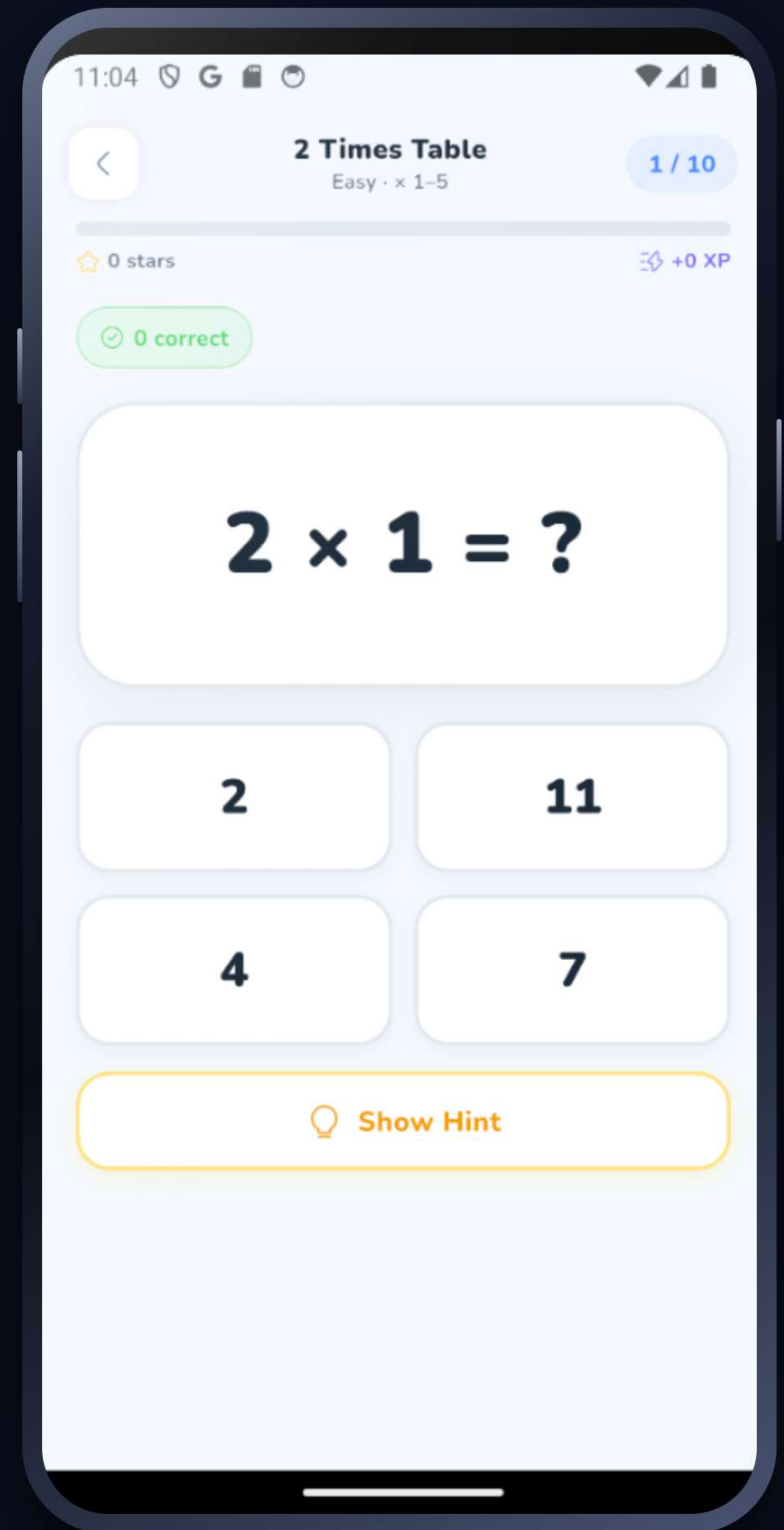
Colour-coded

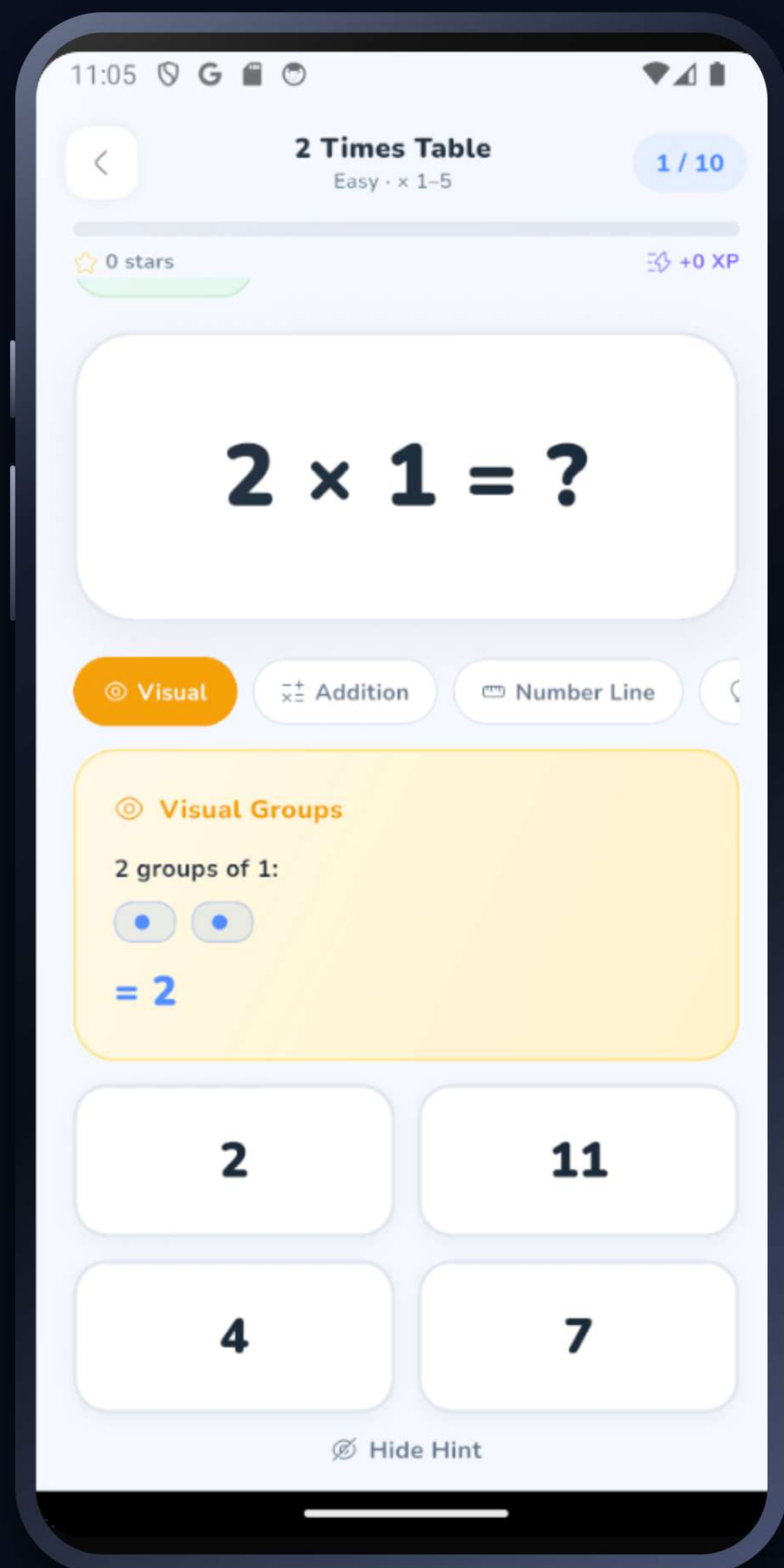
Scannable rows

The lesson engine: one fact at a time

A single question fills the screen. Progress, stars, XP and running accuracy sit at the top; four large answer targets sit below the thumb line.

- Question counter and progress bar
- Live stars, XP and correct count
- Hint available on every question





Four ways to explain the same fact

Children don't share one mental model, so every question carries four switchable hints.

Visual — groups of dots you can count

Addition — repeated addition written out

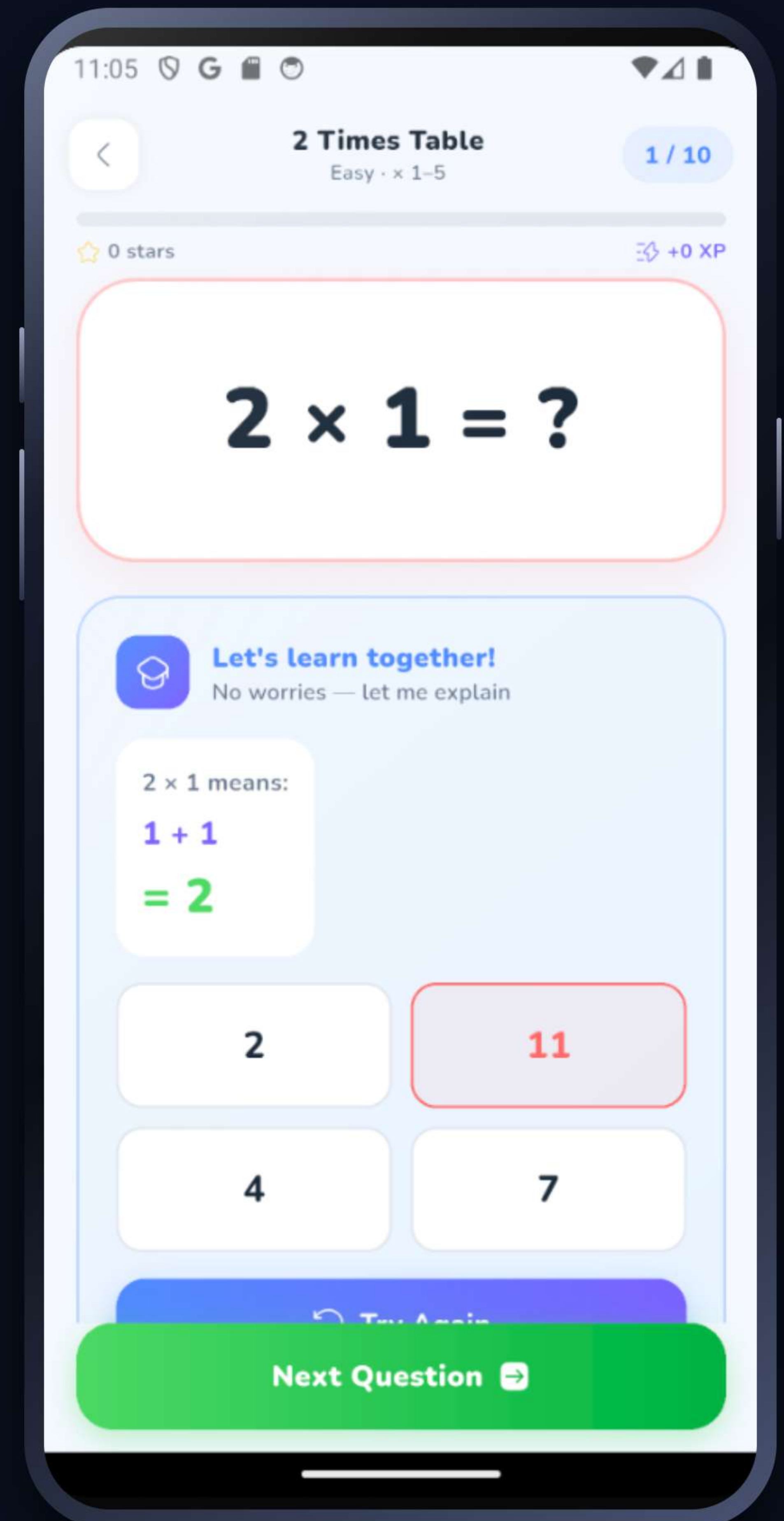
Number line — equal jumps to the answer

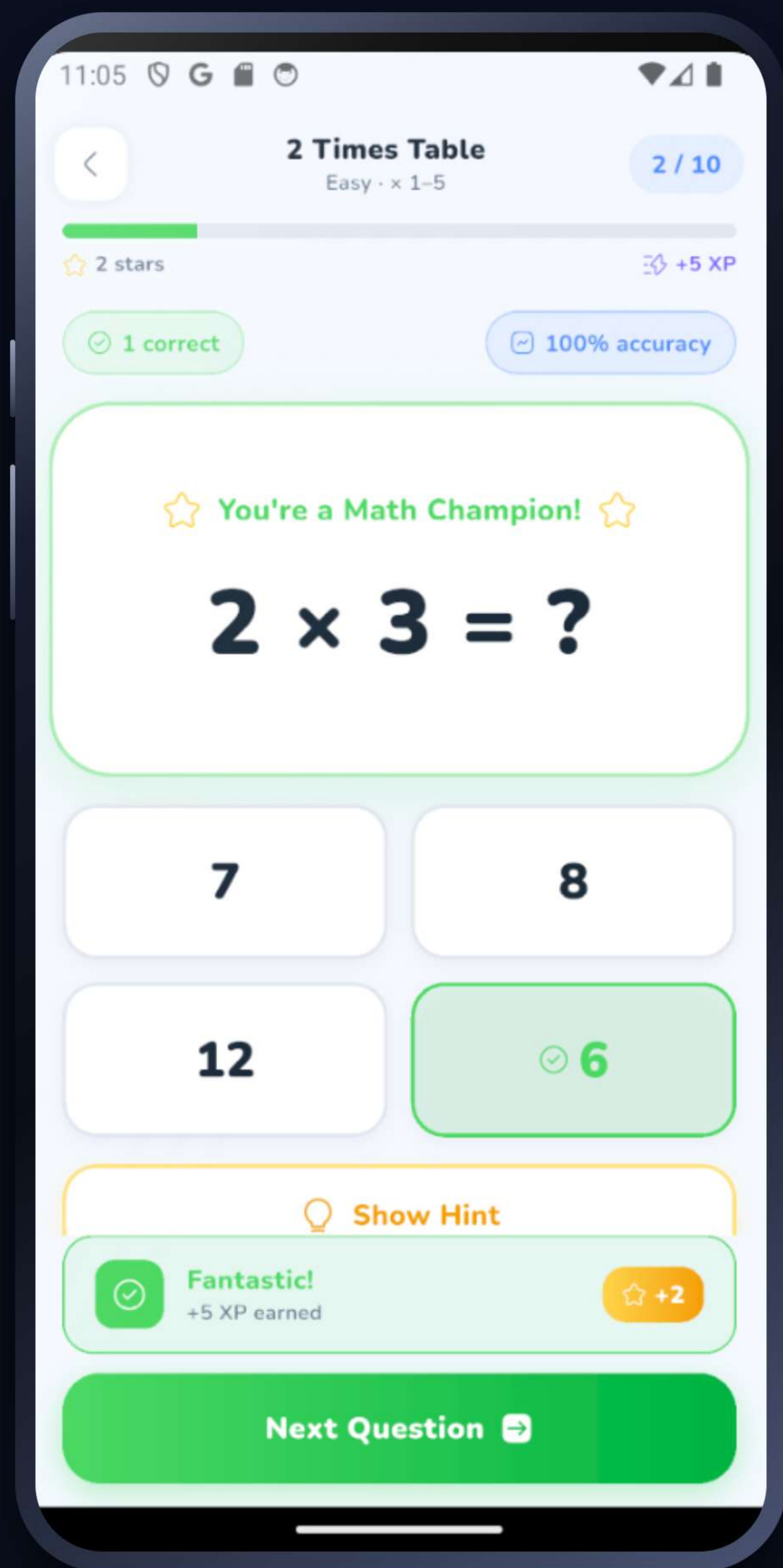
Commutative — flip it to an easier fact

A wrong answer becomes a micro-lesson

No buzzer, no score loss, no "incorrect". The tutor appears, breaks the fact down, and offers Try Again or Next Question the child chooses whether to master it now.

"Let's learn together no worries, let me explain."





Right answers pay out immediately

Green frame, praise line, +5 XP, +2 stars and a live accuracy badge all before the child taps Next Question.

+5 XP

+2 stars

100% accuracy

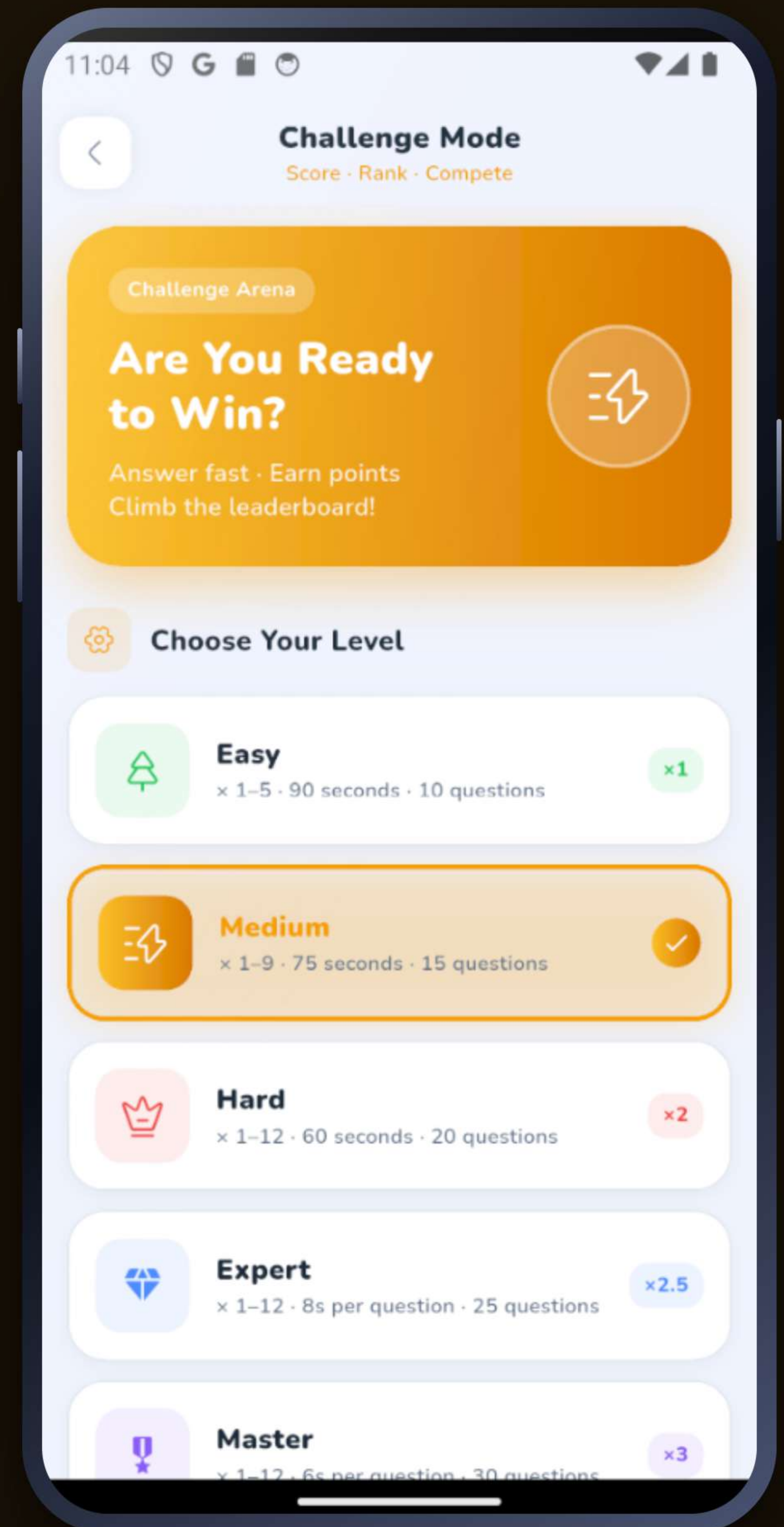
Are you ready to win?

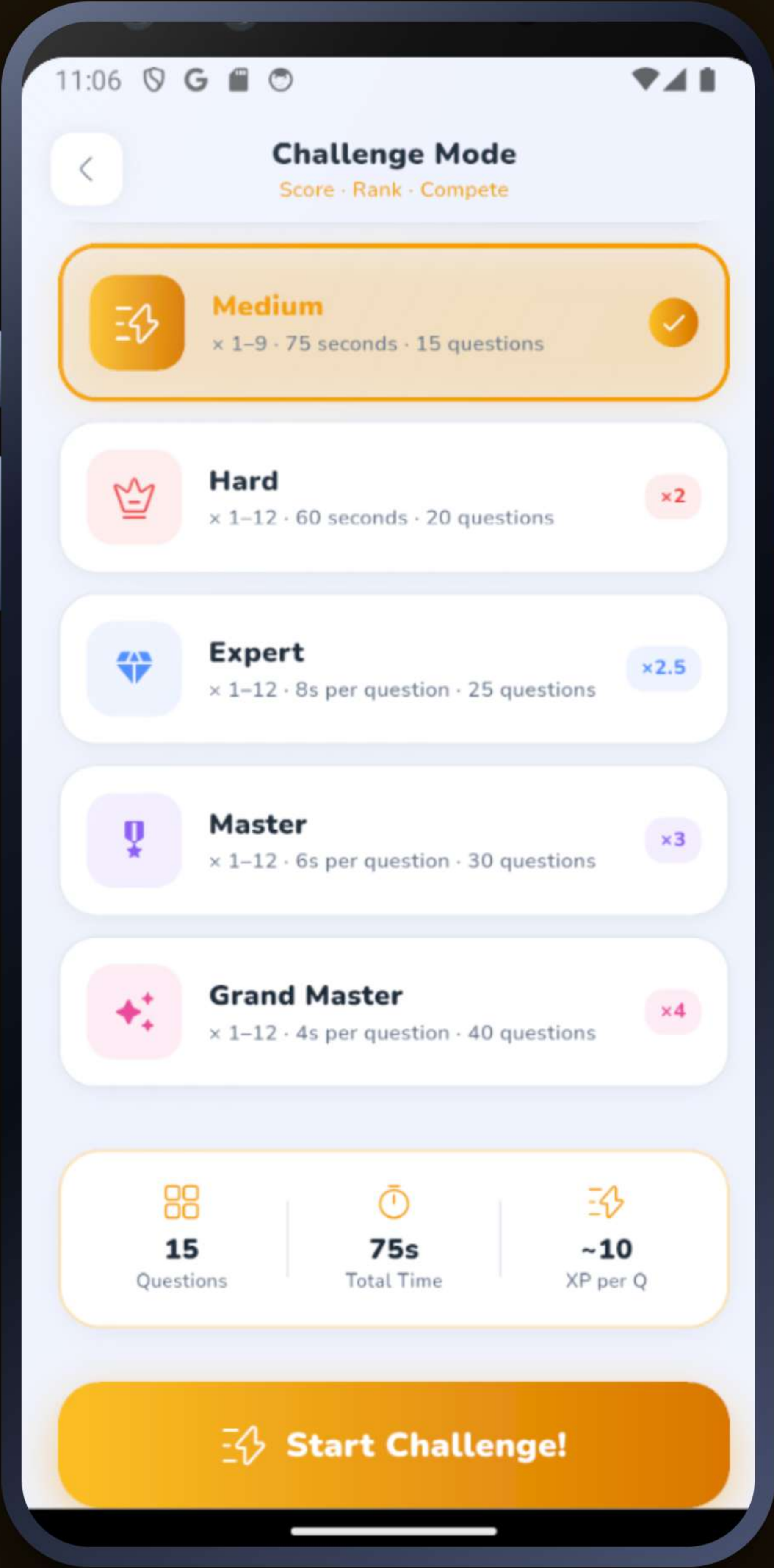
Challenge mode swaps the calm green for arena orange: score, rank and compete. Answer fast, earn points, climb the leaderboard.

Timed

Scored

Ranked





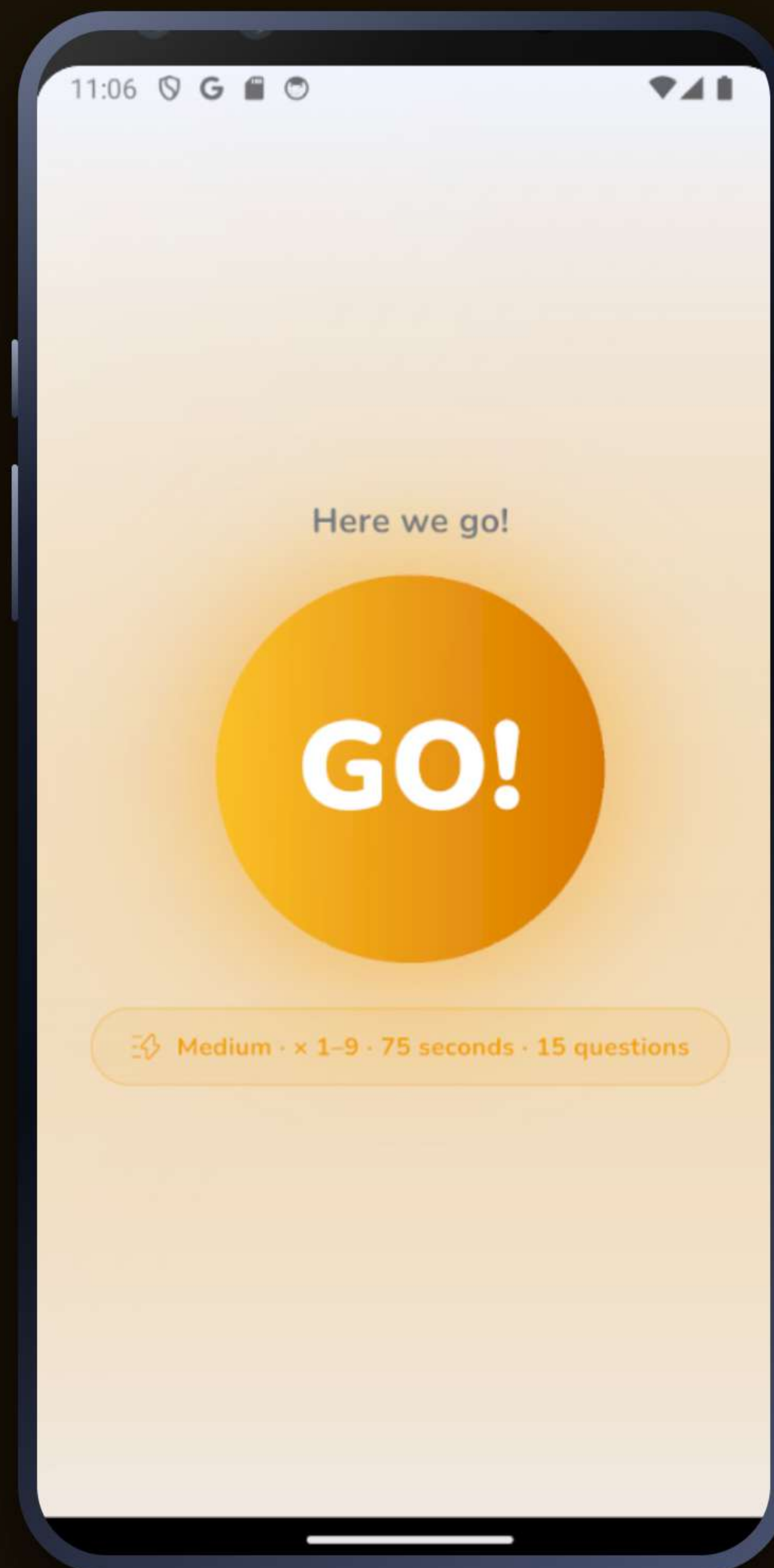
Six tiers, from Easy to Grand Master

Each tier changes the table range, the clock and the score multiplier so the ceiling keeps moving as the child improves.

- Easy x1-5 · 90s · 10 Qs · x1
- Medium x1-9 · 75s · 15 Qs · x1.5
- Hard x1-12 · 60s · 20 Qs · x2
- Expert 8s per question · 25 Qs · x2.5
- Master 6s per question · 30 Qs · x3
- Grand Master 4s per question · 40 Qs · x4

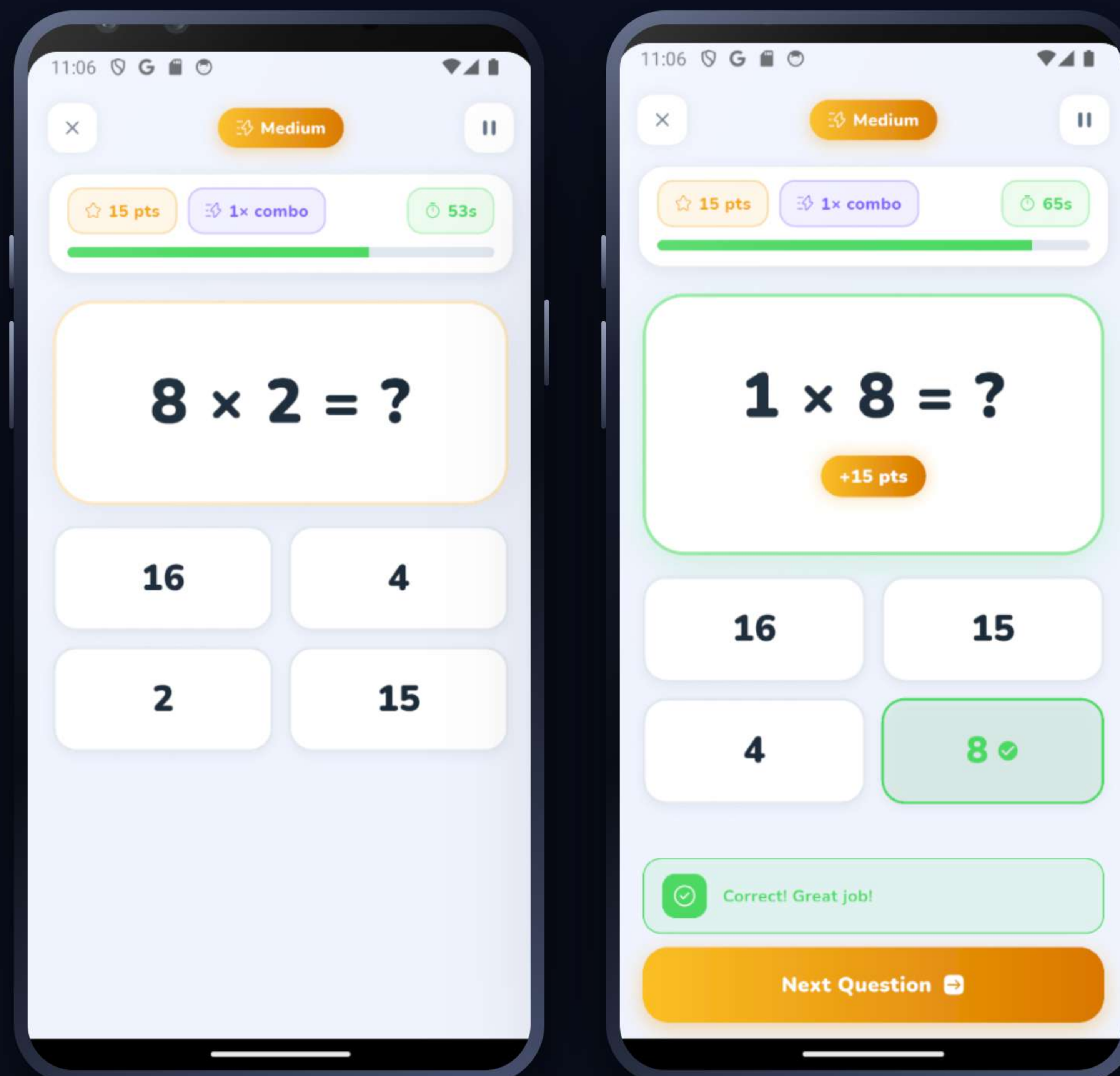
Three seconds of anticipation

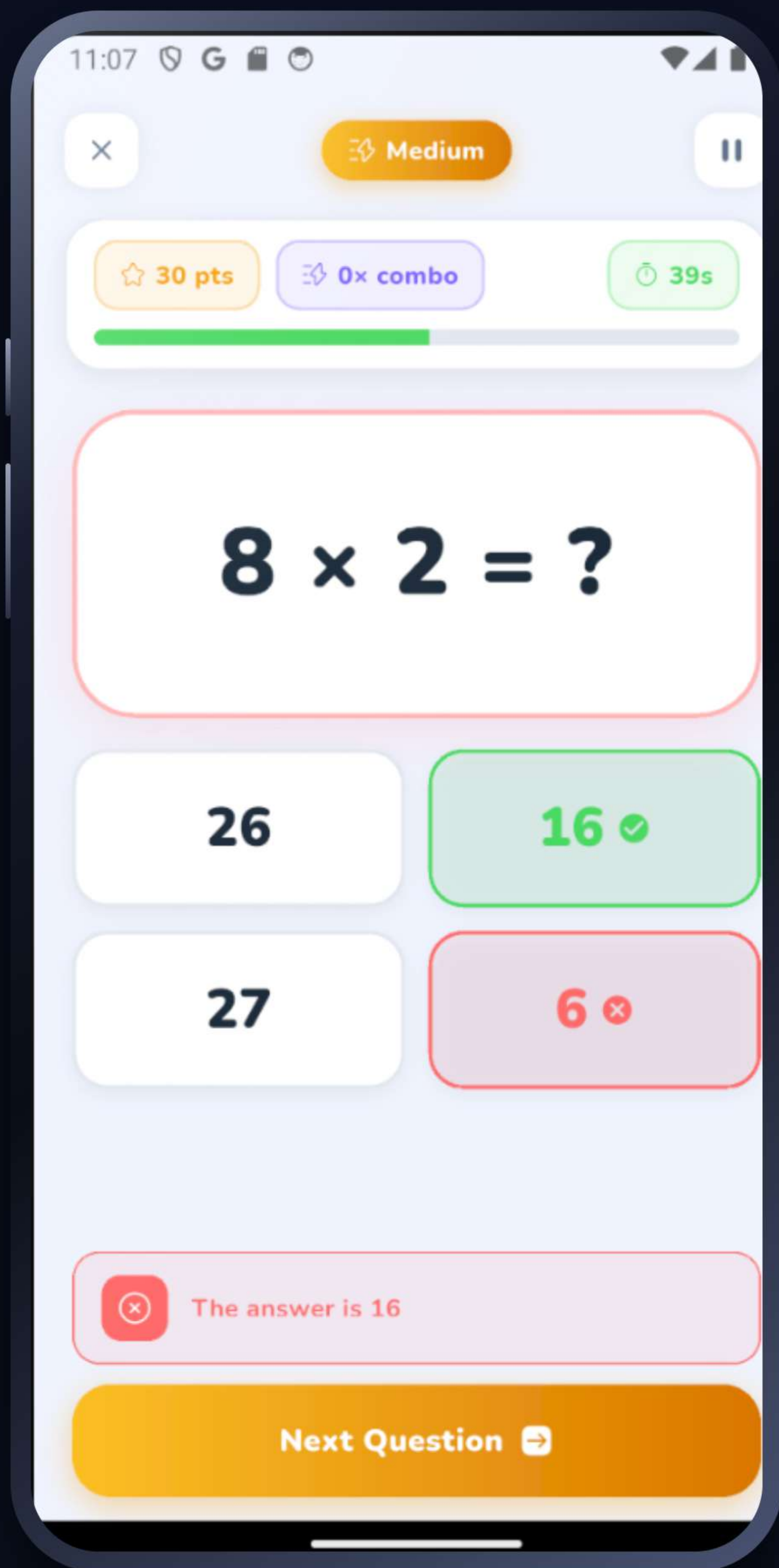
A full-screen countdown between setup and gameplay. It costs three seconds and it makes the round feel like a match, not a form submission.



Points, combo, clock — the whole state in one bar

Score, combo multiplier and remaining seconds live in a single row above the question, with a progress bar draining underneath.





Even under a clock, the answer is revealed

The wrong tap turns red, the correct option turns green, and a single line states the answer. The combo resets to 0x the cost is momentum, never knowledge.

Answer reveal

Combo reset

No score penalty

Challenge complete — every number explained

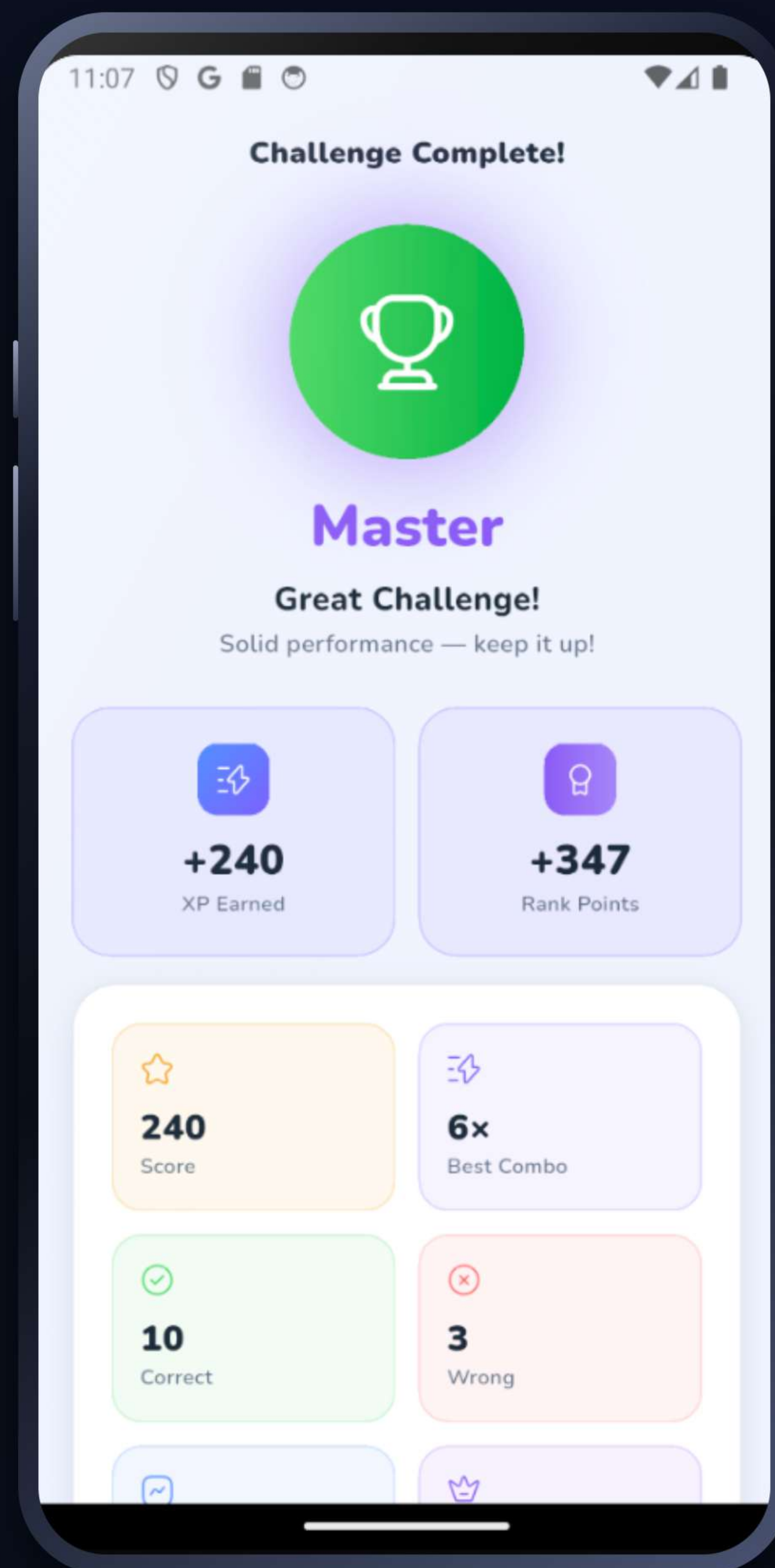
XP earned, rank points, score, best combo, correct and wrong counts, accuracy and tier. One screen a parent can read as easily as a child.

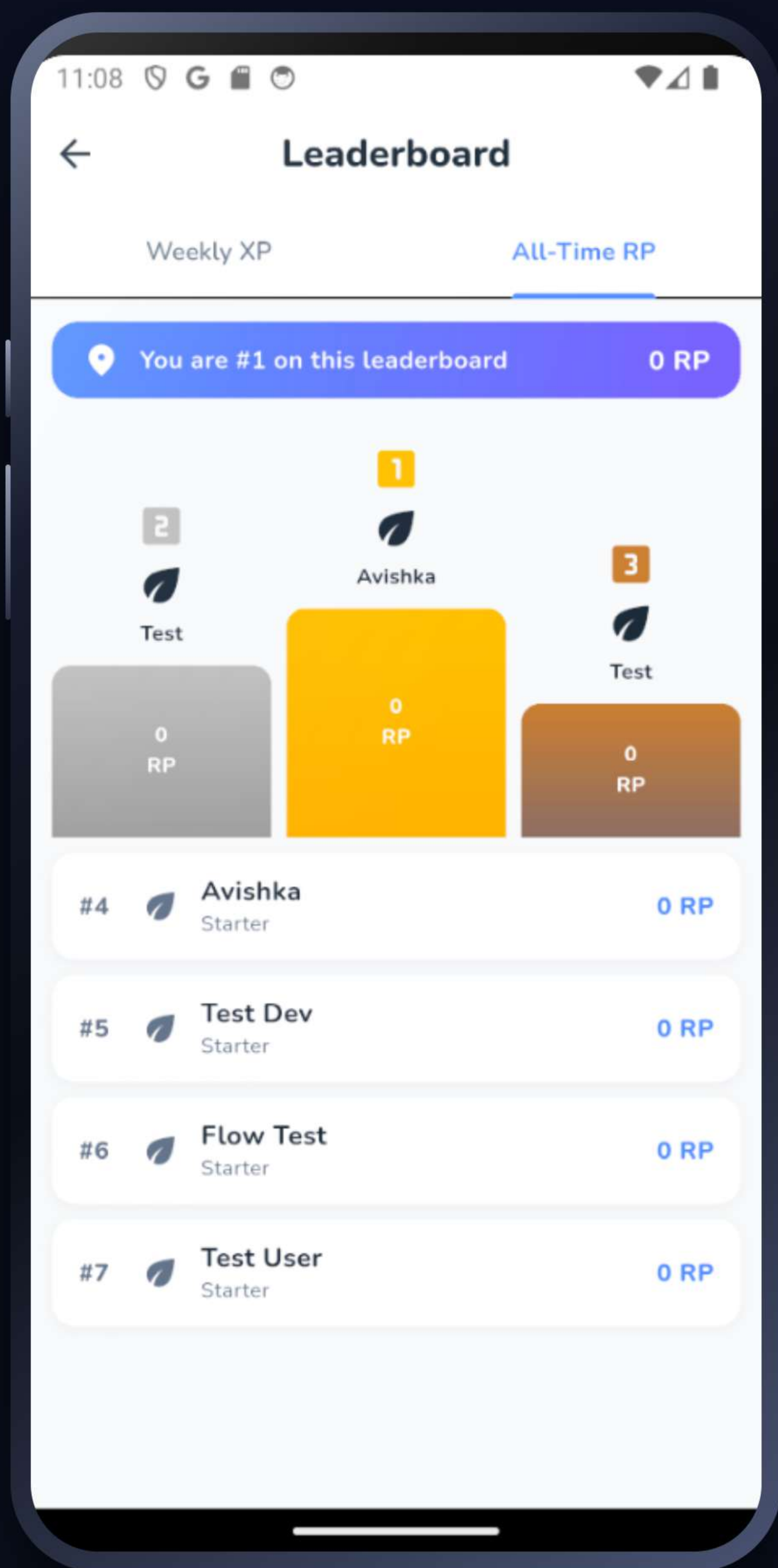
+240 XP earned

+347 rank points

6x best combo

10 correct · 3 wrong





A podium, and your own position pinned

Two boards Weekly XP and All-Time RP with a top-three podium, avatars and rank titles. Your own standing is pinned to the top so you never have to scroll to find yourself.

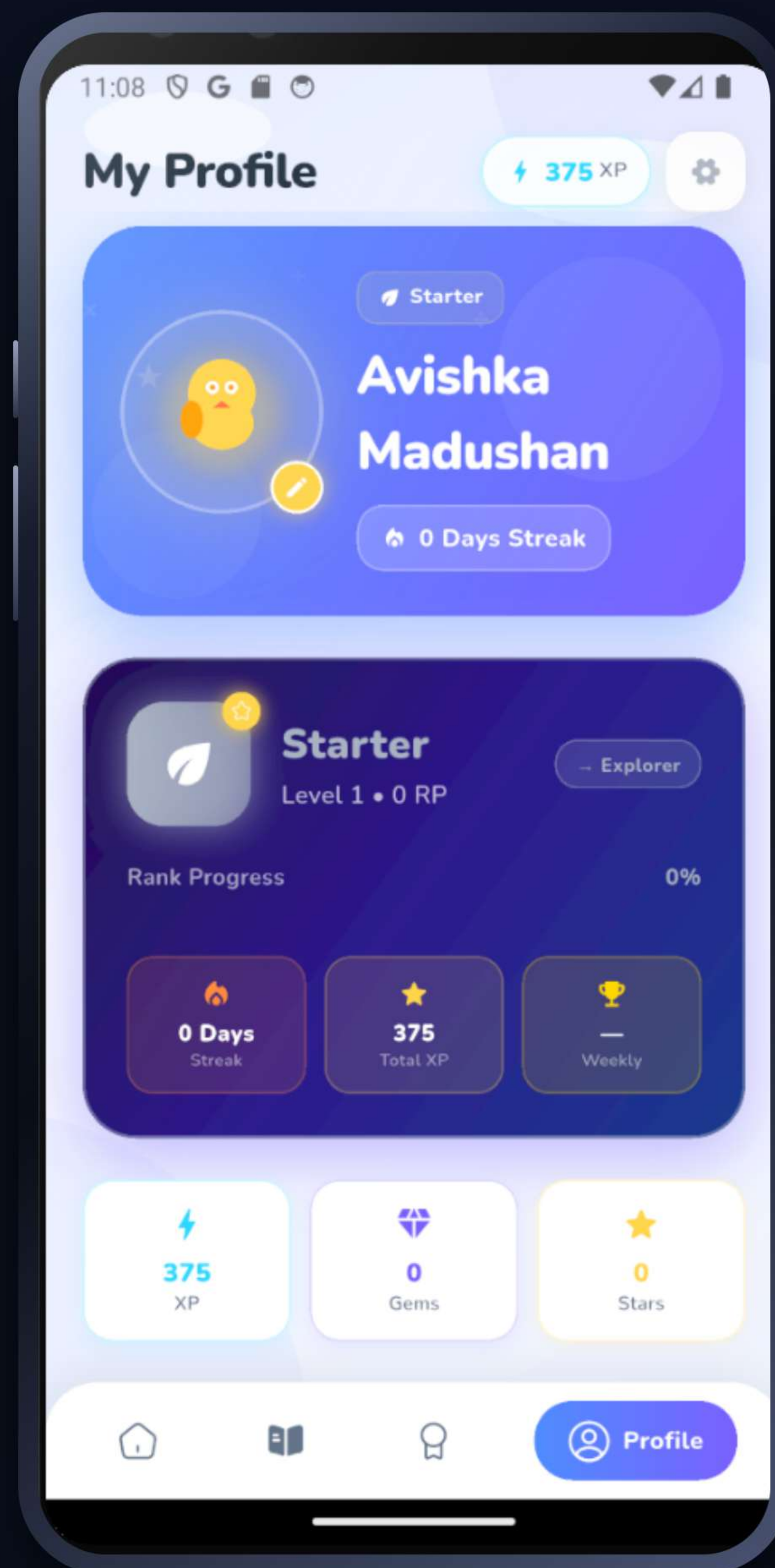
Weekly XP

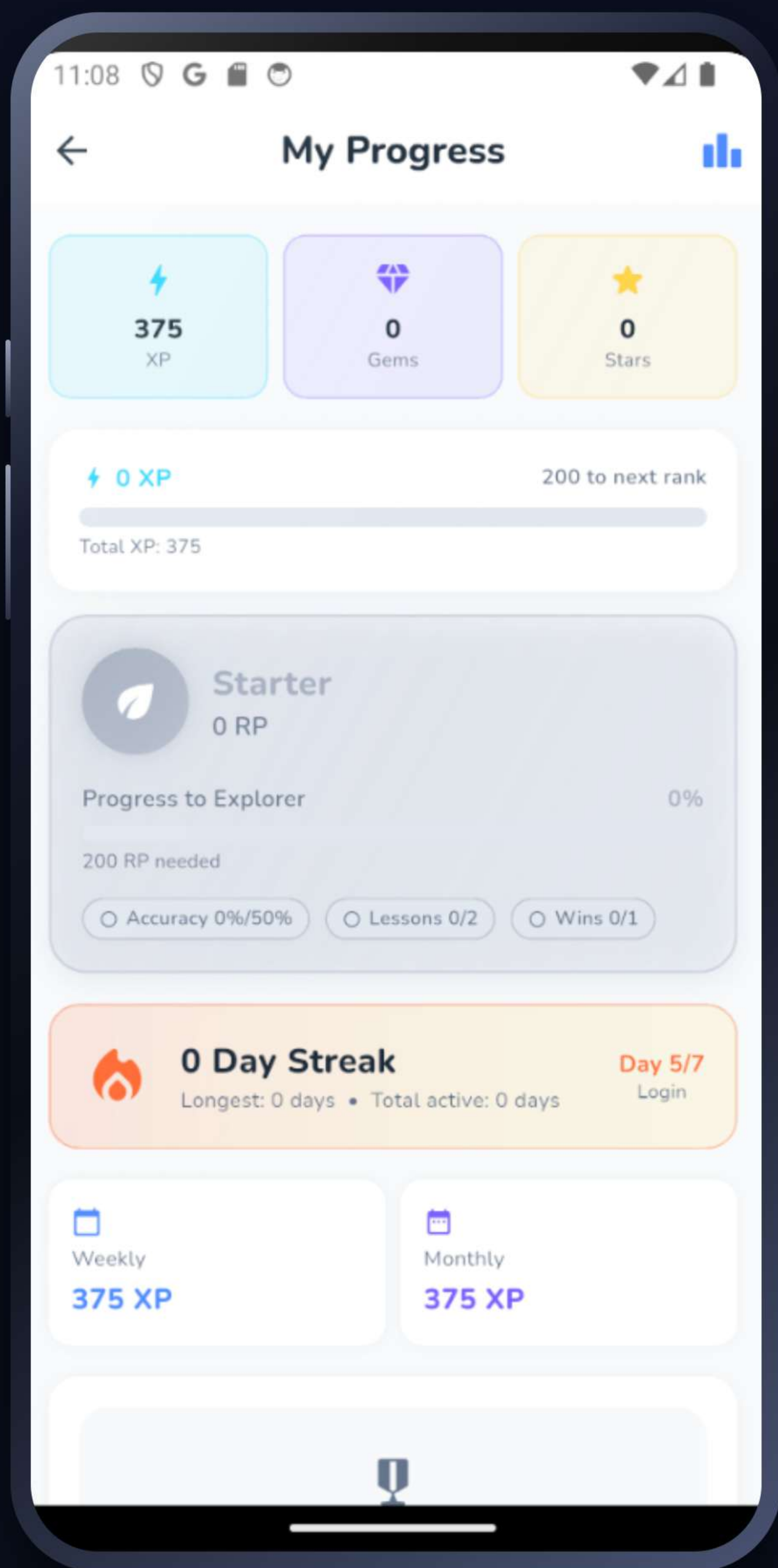
All-time RP

Pinned self-rank

A profile worth decorating

Collectible avatars, current rank and level, streak days, total XP, gems and stars. Avatars unlock with progress and come with their own celebration animation.

375 XP**Gems****Stars**



Progress you can point at

XP to next rank, the exact requirements still open, streak history, weekly and monthly totals, and the badge shelf below.

- Accuracy 0% / 50% target
- Lessons 0 / 2
- Challenge wins 0 / 1

The loop that brings a child back tomorrow

STEP 1

Answer

Earn XP and stars per question

STEP 2

Rank up

Rank points unlock the next tier

STEP 3

Unlock

Badges and collectible avatars

STEP 4

Return

Streaks and local notifications

Compete on the leaderboard → and the loop starts again

Built on one codebase, four targets

CLIENT

Flutter · Dart 3

Android, iOS, Web and Windows from a single feature-first codebase.

API

Node.js · Express

REST endpoints for auth, lessons, progress and gamification data.

DATA

MongoDB

Schema and seed data for users, lessons, ranks, badges and avatars.

OPS

Docker · Nginx · CI/CD

Containerised deployment with Firebase configuration.

google_sign_in

flutter_secure_storage

shared_preferences

lottie

animate_do

shimmer

pinput

flutter_svg

iconsax_flutter

google_fonts

Feature-first, not folder-soup

Every feature owns its own data and presentation layers, so a screen and the service behind it live side by side.

lib/

```
├─ core/ – api client, storage, notifications, theme
├─ features/
│   ├─ authentication/ – login, register, OTP
│   ├─ home/ – dashboard, practice, tables, streak
│   ├─ lessons/ – path, engine, result
│   ├─ challenge_mode/ – timed game, results
│   ├─ achievements/ – badges, gamification service
│   ├─ avatar/ – collection, unlock UI
│   ├─ ranking_system/ – rank card, XP bar
│   └─ leaderboard/
└─ user/ – profile
└─ main.dart
```


Learn. Play. Level Up.

Three taps of the bottom bar: home, challenge, profile.



Five things this project taught me

- 01 Design the reward loop before the screens.
- 02 For children, tone of voice is a feature, not polish.
- 03 One shared engine, many modes, beats four copies.
- 04 Colour can carry mode-switching before any text.
- 05 The unglamorous flows decide whether users stay.

Learn. Play. Level Up!

Mathify a gamified multiplication platform for kids aged 7+.
Flutter client, Node.js + Express API, MongoDB.

Feedback welcome

Open to collaborate